ES112 course plan

(Aug-Nov 2026) (class strength: 380+)

Instructor:

Prof. Shouvick Mondal (*doubt sessions*: by appointment ONLY at AB13/402A)

*There will be **TWO lectures (split into ~1.5 hrs each)**, and **TWO lab sessions (combined into one slot of ~3 hrs)** per week. The L-T-P-C for this course is (3-0-3-4) which means a weekly load of: 3 hrs of lecture, 0 hrs of tutorial, 3 hrs of lab, and overall 4 credits for this course. This course will be offered in both **English (ES 112 (E))** and **हिन्दी (ES 112 (H))**.*

TAs Organization:

TBD

Course email (all announcements will be made): es112_computing-aug-nov-2026.pvtgroup@iitgn.ac.in
Google classroom (all lab assignments will be posted): TBD
GitHub Links:

- https://github.com/SET-IITGN/ES-112-Computing (companion to this course plan)
- https://github.com/SET-IITGN/TracerSET **(useful aid for individuals at all levels and experience)**

ADH Mentors:

TBD

Lectures:

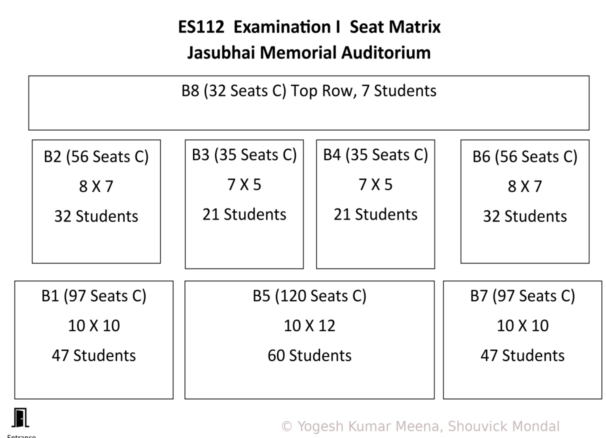
- TBD

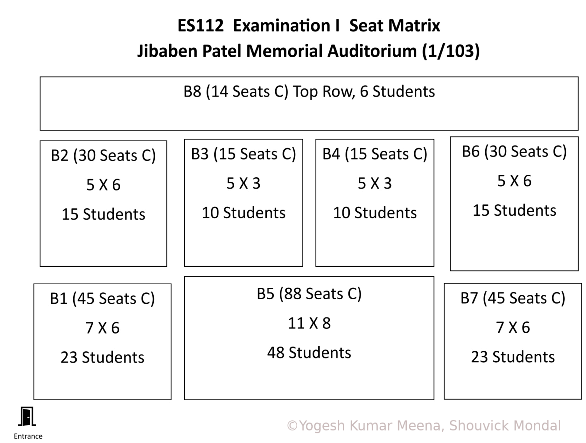
Lab sessions:

- TBD

Evaluation (relative grading):

- [25%] Theory Exam I (TBD)
- [25%] Theory Exam II (TBD)
- [15%] Lab Exam I (TBD)
- [15%] Lab Exam II (TBD)
- [20%] Lab Exam III (TBD)





Previous offerings of this course where I was involved as a co-instructor:

- (public) ES112-Course plan-Aug_Nov_2024.docx
- (public) ES112-Course plan-Aug_Nov_2023.docx

Note #1: There will be no makeup exams, re-exams, or reschedules.
Note #2: Leave requests are not accepted. Standard IITGN norms apply. ***If attendance is below 85%, 'F' grade will be awarded.***
Note #3: There is no prerequisite for this course except sincerity, honesty, and ethics. No usage of mobile phones permitted during lectures and lab sessions. Do not worry if you have zero experience on computer programming, as everything will be taught from scratch, plus the onus is on you to revise the slides before coming to class, and regularly practice programming on a computer system. Any system (w/o Internet) with Python >=3.10 installed will be sufficient for the lab experiments. Note that the computer labs at IITGN have a dedicated system (desktop) per student. However, you may bring your own systems if you wish. Lab sessions will take place online on HackerRank (web application) wherein your account must be associated with your <rollno>@iitgn.ac.in email. Any other account will be ignored. The usage of the online mode is recommended as it does not need you to locally install Python as the web application is integrated with the necessary and sufficient software required for the experiments. However, when using a computer system belonging to the lab, make sure to do a proper logout from your HackerRank account. Otherwise, the next person using that system will automatically bypass login and overwrite your work which cannot be recovered. The onus is on you to make sure this is taken care of.

Overall course load:

TBD

Timeline:

Excluding holidays and breaks. **?** *indicates lab slots consumed by institute-reschedules/holidays/exams/breaks.*

Month	**Day**	**Topics (slides)**
August		
September		
October		
November		

Course Contents:

(tentative and subject to change at the sole discretion of the instructor)

1. *Introduction to computing*: computer hardware architecture, interpreter versus compiler, writing your first program, syntax error, debugging;
2. *Language basics:* keywords, data-types, variables, expressions, statements, operator precedence and associativity, static scope and its nesting, user input;
3. *Conditional execution*: boolean expressions, logical operators, exception handling, flowcharts for conditional constructs;
4. *Iterative execution:* loops, speed of convergence, usage of break and continue, nested loops, flowcharts for iterative constructs;
5. *Strings*: string as a sequence of characters, traversal using a loop, slicing, immutability, parsing, comparison, formatting;
6. *Lists*: homogeneous versus heterogeneous lists, traversal, indexing, addition, deletion, slicing, aliasing, comprehension, nesting;
7. *Functions*: built-in versus user-defined, nested functions, call versus return, flowchart for interprocedural execution, global variables,
8. *Recursion:* recursive definitions, visualizing recursive computations, direct versus indirect, finite versus infinite;
9. *Files:* persistent storage, operations: open, read, write, seek, tell, etc.;
10. *Dictionaries*: ordering, mutability, homogeneity, sliceability, and indexing strategy, dictionaries and files, looping and dictionaries;
11. *Tuples:* tuples as immutable lists, assignment, packing, unpacking;
12. *Object oriented programming*: using objects, classes as types, object lifecycle, inheritance, overriding;

13. *The Python Standard Library*: built-in functions, built-in constants, built-in types, built-in exceptions, text processing services, more datatypes, numeric and mathematical modules, functional programming modules, operating system services.

Student Honour Code: [https://iitgn.ac.in/students/honourcode]

TextBook:

1. Charles Russell Severance, Sue Blumenberg, and Elliott Hauser. 2016. *Python for Everybody: Exploring Data in Python 3*. CreateSpace Independent Publishing Platform, North Charleston, SC, USA. Free: https://www.py4e.com/book
2. Allen B Downey. 2024. *Think Python: How to Think Like a Computer Scientist.* (3rd Ed.) Green Tea Press. Free: Think Python

References:

3. Eric Matthes. 2023. *Python Crash Course. A hands-on, project-based introduction to programming* (3rd edition). No Starch Press.
4. Al Sweigart. 2025. *Automate the Boring Stuff with Python: Practical Programming for Total Beginners* (3rd ed.). No Starch Press, USA. Free: Automate the Boring Stuff with Python
5. The official documentation for Python 3 (https://docs.python.org/3/index.html)

References *(for experts)*:

6. Python's Compiler Design (https://github.com/python/cpython/blob/main/InternalDocs/compiler.md)
7. Guide to Python's Parser (https://github.com/python/cpython/blob/main/InternalDocs/parser.md)
8. PEP 617 – New PEG parser for CPython (https://peps.python.org/pep-0617)
9. Packrat parser (https://en.wikipedia.org/wiki/Packrat_parser)

Software Tool Support *(command line)*:

10. `trace` (https://docs.python.org/3/library/trace.html)
11. `snoop` (https://github.com/alexmojaki/snoop)
12. `pysnooper` (https://github.com/cool-RR/PySnooper)
13. `pdb` (https://docs.python.org/3/library/pdb.html)
14. `tokenize` (https://docs.python.org/3/library/tokenize.html)
15. `ast` (https://docs.python.org/3/library/ast.html)
16. `dis` (https://docs.python.org/3/library/dis.html)

Learning Outcomes:

At the end of this course, students: (i) will learn to communicate with computers in at least one major general purpose programming language relevant in today's world, (ii) will have the necessary background for forthcoming programming related courses in their curriculum, (iii) with sustained practice will be prepared to undertake coding sessions in job interviews and/or public competitions.

Other Remarks:

This UG core course is meant only for B.Tech students and serves as the first course on computer programming (computing). The foundational concepts will be useful for problem-solving through systematic instructions provided to computers in Python.

"*Python is currently the most popular programming language worldwide, maintaining a lead of more than 10 percentage points over its closest competitors.*" – TIOBE Index for February 2026. The TIOBE Programming Community index is an indicator of the popularity of programming languages.